Market Experiment & Project Management

Week 7 Product Launch Management

MKT 3717 Product and Brand Management

Dr. Dayoung Kim

Today

- Designing “Market Experiment” to predict the success of product
  - Motivation: Building Causality
    - A/B testing

- Project Management
  - Defining and Organizing Projects
  - Constructing Project Networks
    - Defining the Work Breakdown Structure (WBS)
    - Diagramming the Network
    - Developing the schedule using critical paths

6. Analytic Approaches for Market Testing

Motivation: Causality

For any marketing effort, we almost always have causal relationships in mind!

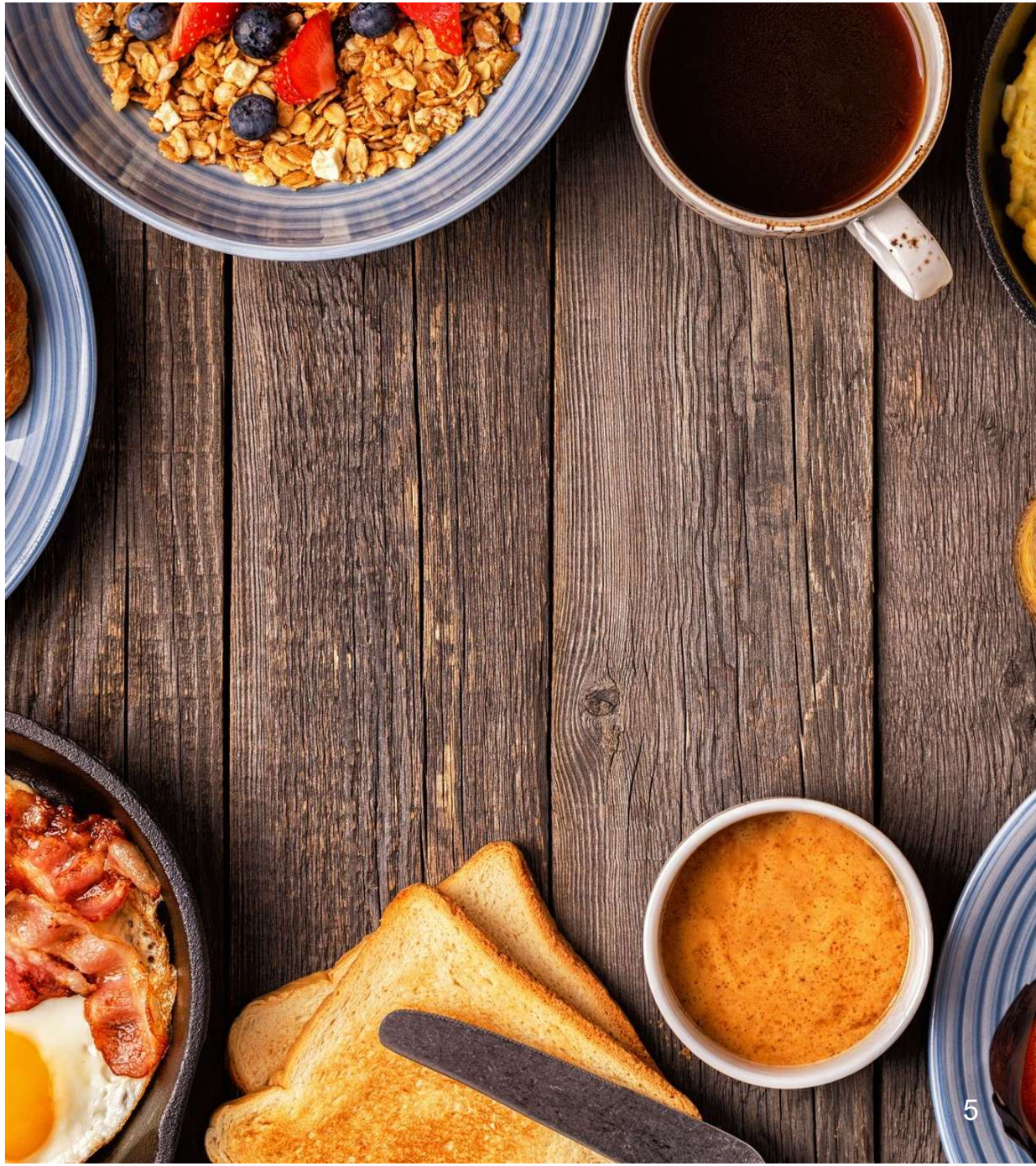
- Does this new product design improve user satisfaction?
  - *What is the causal effect of the new design on user experience? Are users more satisfied because of the new design or because of some external factors?*
- Is this marketing program going to work?
  - *What is the causal effect of the new marketing program on customers' decision-making? Is it positive? Is it worth the investment?*

Establishing causality

- Establishing a causal relationship is very difficult in reality

Thought experiment: Should I eat breakfast before I take an exam?

→ In an ideal world (assume you are all-powerful), how would you identify the causal <u>effect of eating breakfast on the exam performance</u>?



Practical solutions

- We (sadly!) cannot “replay” the world in different scenarios
  - The realized world is called factual scenario
  - A hypothetical world (that did not occur) is called counterfactual scenario

- We have two practical solutions to infer causality in the real world
  - Experiments (randomized controlled trials)
  - Econometric (statistical) analyses of observed data

Practical solutions

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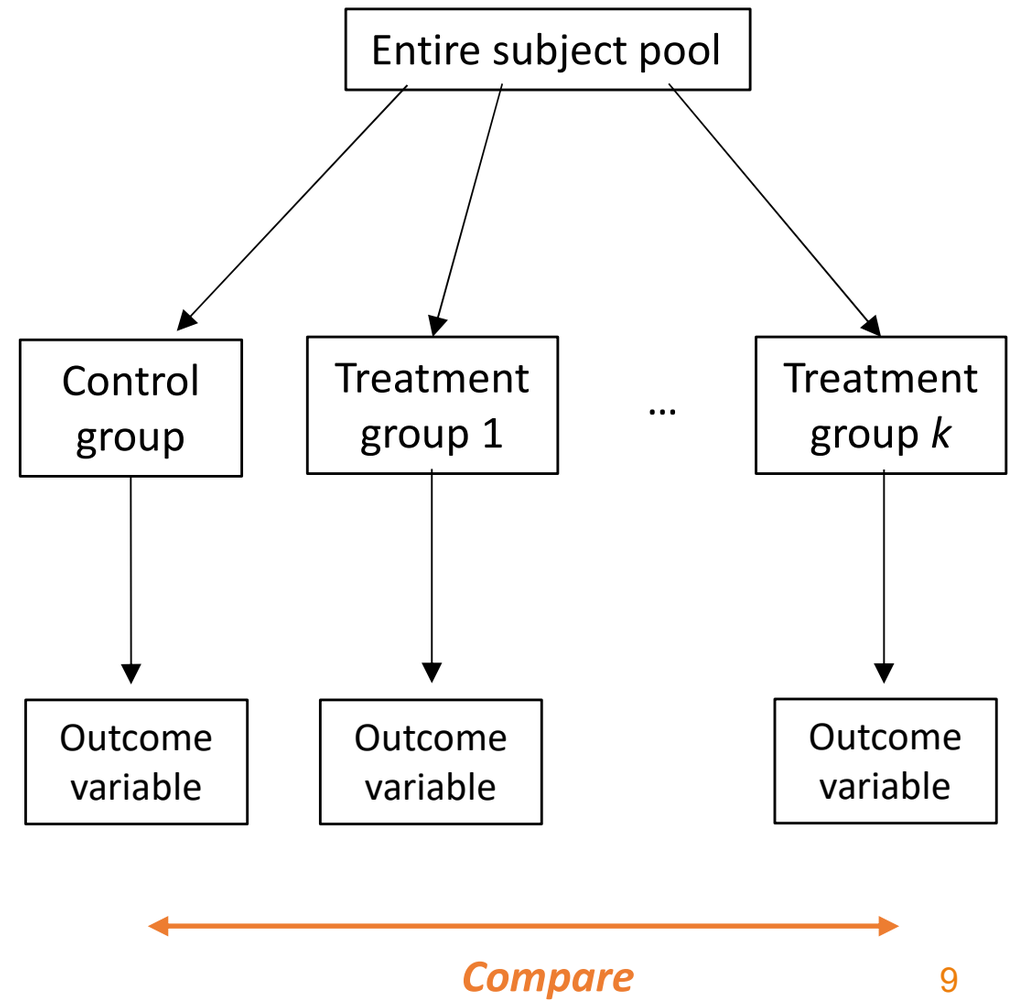
Practical solutions

Experiments (Randomized Controlled Trials)	Data analysis of observational data
• "Gold standard" • We randomly assign subjects into different groups (scenarios), and compare outcomes • Randomization ensures that, in expectation, the only difference between groups is the treatment variable we are manipulating • Sometimes experiments are not practically feasible due to time constraints, ethical constraints, monetary constraints, etc.	• We attempt to get "close" to establishing causal relationships by investigating associations between observable variables • Example: Regression analysis • Some advanced methods can get us pretty close to causality • Caution necessary when interpreting results (association vs. causation)

Experiments

1. **Randomize** subjects into different treatment groups
2. In each group, we manipulate the explanatory variable of interest
   - *Everything else must be the same across groups*
3. Let the experiments run
4. We compare the average of the outcome variable between treatment groups



Experiments (2)

- In an ideal experiment, the only thing that differs between groups is the explanatory variable that we manipulated
  - Everything else is, in expectation, identical
  - Note: If you can go back in time, everything else <u>is</u> identical
- Therefore, if you observe any differences in the outcome variable, we can say this is caused by the explanatory variable we manipulated

Simple Example

- Suppose you want to test the hypothesis that Apple Watch helps reduce people's weight
  - X = Indicator variable for whether you own an Apple Watch or not
  - Y = Weight
- You surveyed people to obtain data for X and Y, and the two variables were strongly negatively correlated.

(d) Rich people are likely to own an Apple Watch, and weigh less

(b) Overweight people trying to lose weight may be more likely to buy Apple Watch

(a) Apple Watch reduces weight

(c) Fit and health-conscious people may be more likely to buy Apple Watch

(e) Men are more likely to own an Apple Watch, and weigh more

Simple Example (continued)

- (a) is the treatment effect we want to estimate
- (b), (c), (d), (e) are confounding factors

- Correlation simply shows a net statistical association that accounts for effects (a), (b), (c), (d), (e) as well as all other underlying confounding factors that exist

- It is therefore difficult to infer the effect size of (a), the impact of Apple Watch on weight

Simple Example (continued)

- Experiments can resolve this problem through **randomization**:
  - Recruit subjects who do not yet own an Apple Watch
  - **Randomly** give Apple Watch to 50% of the subjects
  - Measure their weight (e.g., after two months)

- Randomization helps ensure that, in expectation, the only difference between the treated (received Apple Watch) and the control (did not receive Apple Watch) groups is the provision of Apple Watch (treatment variable you manipulated)
- Hence, any difference we observe in weight two months later can be attributed to the provision of Apple Watch

A/B Testing

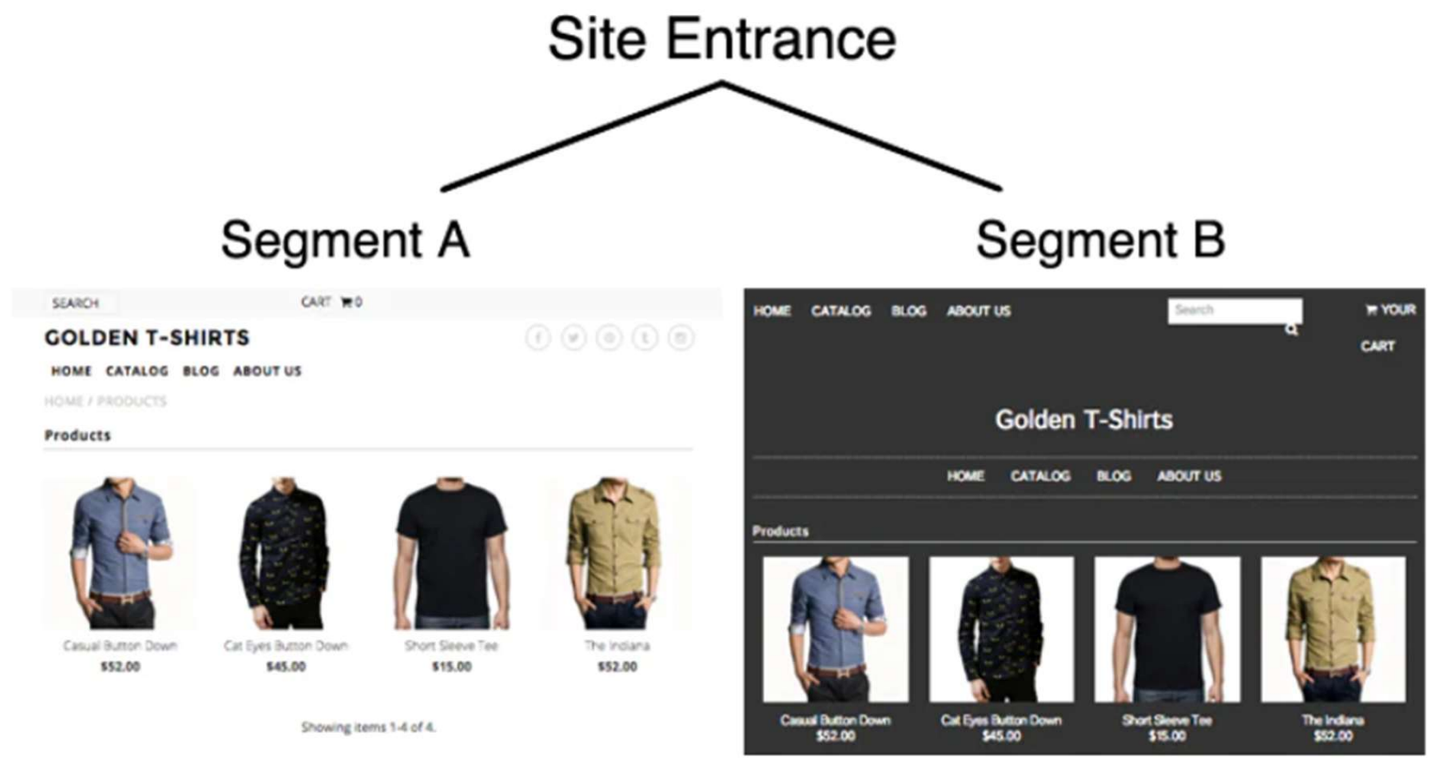
- A/B testing is a simplest form of experiment in which you test the effectiveness of two competing options, A and B

1. Randomly assign participants into A (control) and B (treatment)
2. Measure the outcome variable(s) of interest
3. Compare the outcome variable(s) between A and B

Useful in testing various marketing tactics!

A/B Testing Example (1) Webpage design

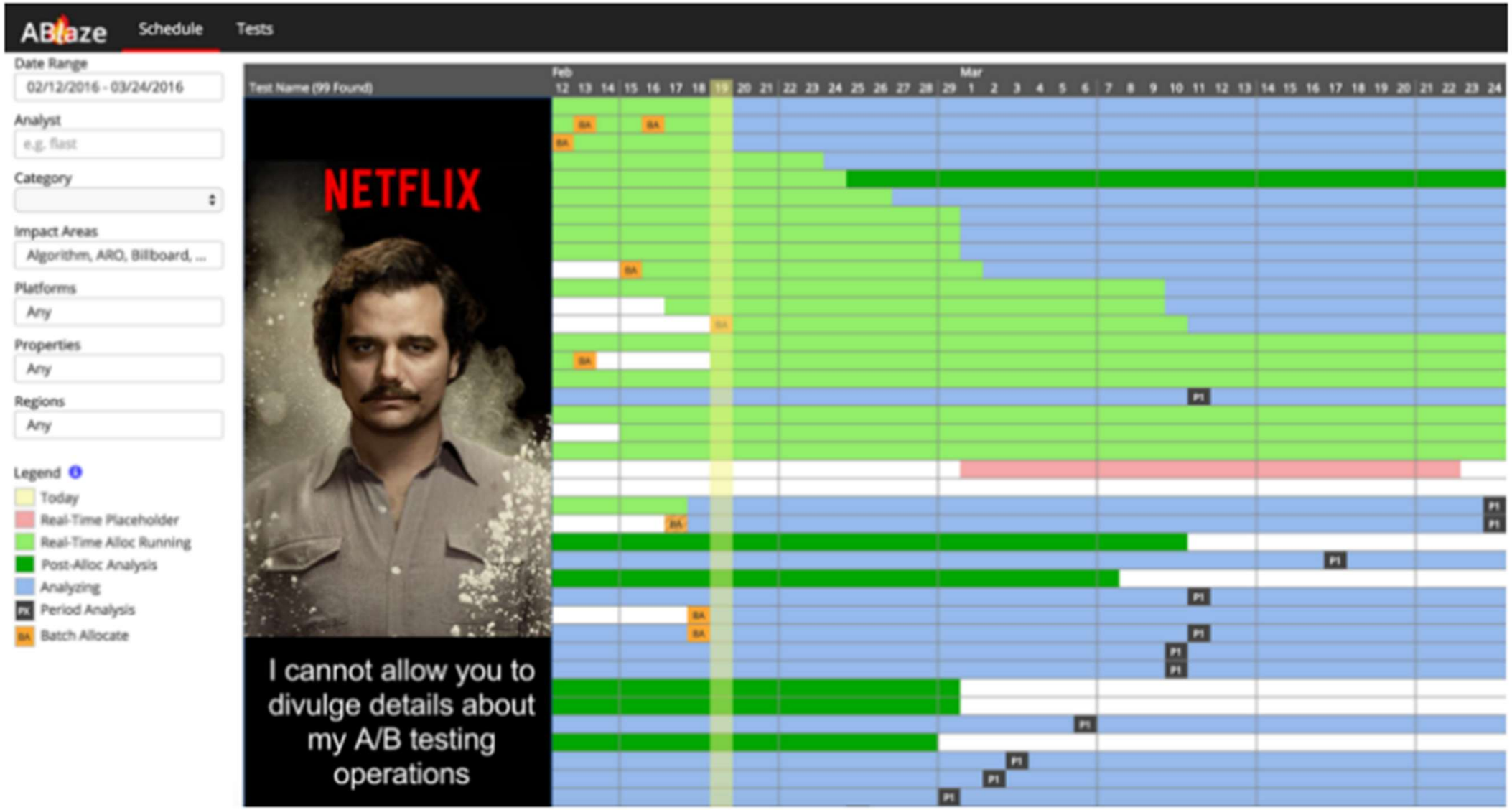
- Post launch test for webpage UI design



A/B Testing Example (2) Netflix UI

Cells	Cell 1 (Control)	Cell 2	Cell 3
Box Art	Default artwork		

- Images associated to movie titles, sometimes resulting in 20-30% more viewing



- Allocation of subjects should be carefully done (to avoid conflicting tests)

Factorial Design

- Often, you want to test more than two options at the same time, and/or test more than one dimension at the same time

- For example, you want to find the optimum combination of text information (first dimension) and product suggestion (second dimension) in a personalized email ad for wine:

	Short text	Medium text (includes taste profile)	Long text (includes taste profile + food pairing)
$24 Cabernet Sauvignon			
$32 Rioja			

- In the above example, there are 2 (product) x 3 (text information) = 6 conditions
- This is called a **factorial design**

Factorial Design (2)

- Factorial design can reveal **optimal combination** of multiple variables

- For example, table below shows the profitability of each condition:

	Short text	**Medium text (includes taste profile)**	**Long text (includes taste profile + food pairing)**
$24 Cabernet Sauvignon	Medium	Low	Low
$32 Rioja	Low	Low	High

- If you iterate A/B testing, starting from the first cell [short text & $24 Cabernet Sauvignon], you may never reach the most profitable combination of [long text & $32 Rioja].

Between subjects vs. within subject

- There are two ways to implement an experiment:
  - **Between subjects**: Each subject receives *one* random condition
  - **Within subjects**: Each subject receives *all* conditions

- Between-subjects design is most common. Within-subjects design has pros and cons, and can be a good alternative in some circumstances:

Pros of within-subject design	Cons of within-subject design

Key: Random Assignment

- Why are experiments considered the "gold standard"?

- Random assignment helps control for all confounding variables
  - Ensures that the only difference (in expectation) across the different conditions is the treatment variable that you are manipulating
  - Confounding variables are variables that affect both the treatment and dependent variables

- This allows you to infer a **clean causal relationship** between X (treatment variable) and Y (dependent variable).

Endowment Effect

- **Endowment effect** refers to a phenomenon where people tend to value an object higher if they own it than if they do not own it

- Mug experiment by Kahneman, Knetsch and Thaler (1990):
  - Half the class received a mug, which sold for $6 each
  - Students bid for the mug
  - Median owner: Unwilling to sell for less than $5.25
  - Median buyer: Unwilling to buy for more than $2.50

Reference: Kahneman, D., Knetsch, J. L., & Thaler, R. H. (1990). Experimental tests of the endowment effect and the Coase theorem. *Journal of Political Economy*, *98*(6), 1325-1348.

Class Experiment: Endowment effect with Lindor chocolate?

1. I randomized you into two groups
2. Endowed (treatment) group received a chocolate
3. I let you fill out a valuation survey
4. Let’s compare the mean of the outcome variable

Statistical Significance

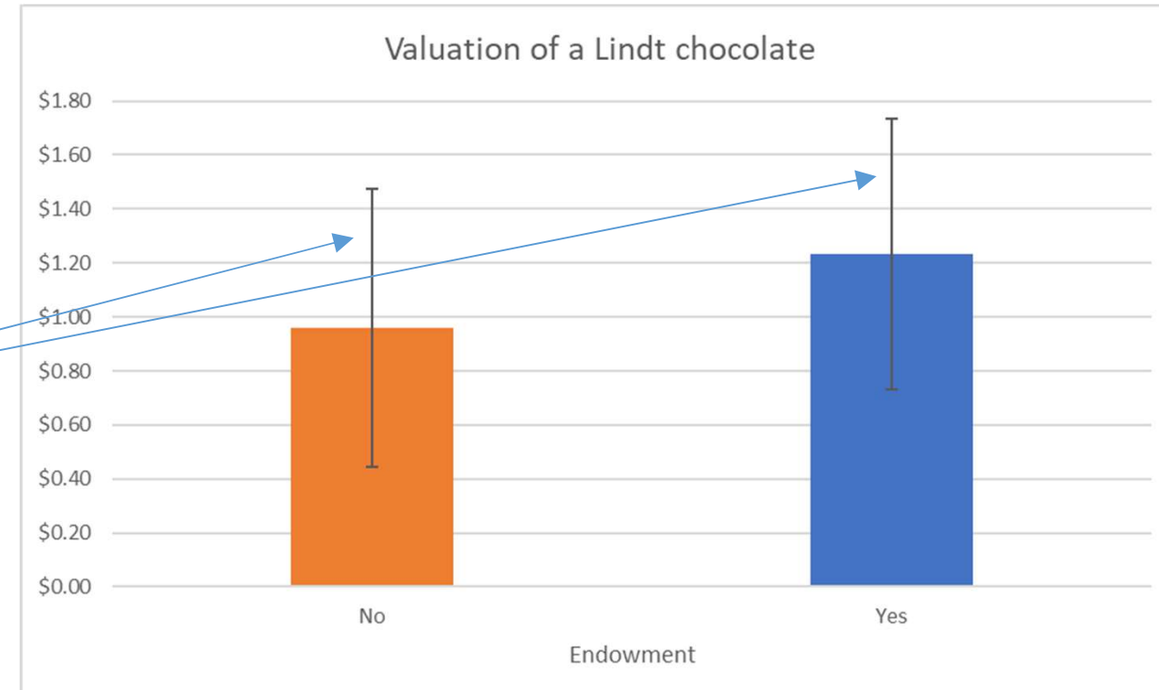
- We say that a difference is statistically significant if it is unlikely to have occurred by chance

Two ways to examine statistical significance:

A. 95% confidence intervals
   - Visually more appealing
B. Regression analysis
   - Exact test with p-value

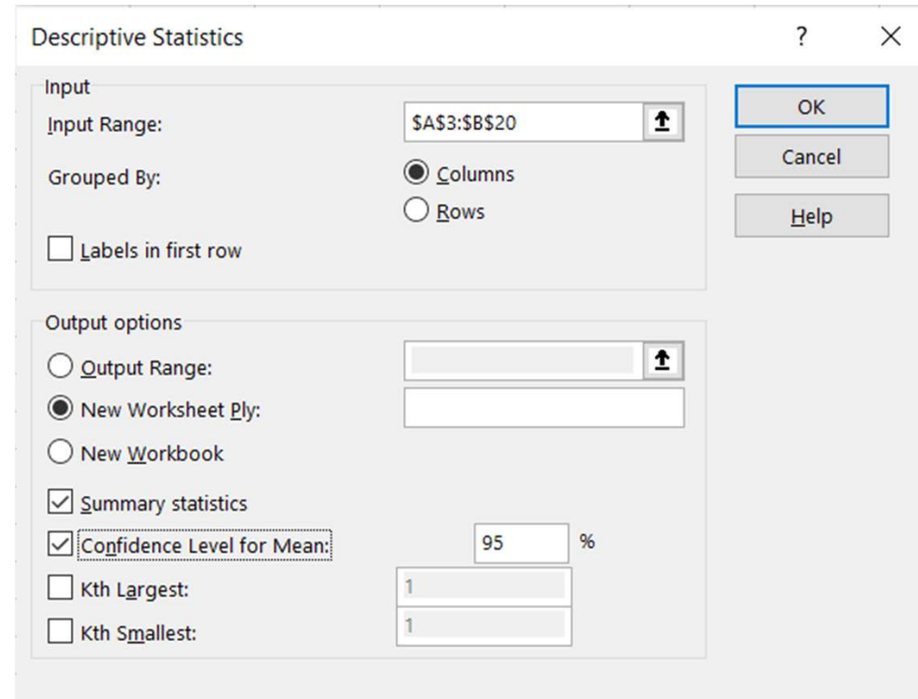
A. 95% confidence intervals

- The means we obtained are sample means, i.e., means of a random sample
  - The means would change a little if a different random sample was obtained
- The sample means approximate true population means, which is what we are interested in
- Confidence intervals (CI) show the range in which we are confident that the true population mean lies within
- Question: How should we interpret the results?
  - If CI's overlap, difference is not statistically significant
  - If CI's do not overlap, the difference is statistically significant



Computing 95% CI in Excel

- Data -> Data Analysis
- Select: Descriptive Statistics
- Input range: Select data entries
- Check: Summary statistics
- Check: Confidence Level for Mean: 95%



- Mean + Confidence Level(95%) is the upper bound of interval
- Mean – Confidence Level(95%) is the lower bound of interval

B. Regression analysis (Recap)

- Another way to test for the difference is with regression analysis:

$$Y = a + b\,X$$

Outcome variable

Here, it is chocolate valuation ($)

Coefficients (Parameters)

Explanatory variable

Here, it is endowment (0 or 1)

- Coefficient b is the coefficient of interest that shows the relationship between Y and X.
- Statistical significance of b *will be shown by p-value (significant if sufficiently small)*

Regression analysis in Excel

- Let's estimate the following regression equation:

$$\text{Value}_i = a + b\ \text{Endowed}_i$$

  - Value_i: Subject i's valuation of chocolate in dollars
  - Endowed_i: Whether the subject was endowed (1) or not (0)

- Then,
  - Mean valuation of not-endowed group: a
  - Mean valuation of endowed group: a + b
  - b: Endowment effect

B. Regression analysis (Recap)

- Regression analysis gets us closer to establish causality, but it is not perfect
  - Infinitely many unobservable variables
  - Causal direction is assumed, not tested

- More advanced techniques can get us even closer (beyond scope of this course)
  - Difference-in-differences
  - Regression discontinuity
  - Propensity score matching

Summary: Steps to designing an experiment

1. Do background research using existing data
   - Descriptive analytics
   - Cluster analysis
   - Regression (Next week..)
2. Form hypotheses and ideas that you want to test
   - Prioritize your ideas
3. Design the experiment
   - How to measure the outcome variable of interest
   - Unit of randomization
   - Number of treatments
   - Between- vs. within-subject design
   - Incentive compatibility check
4. Run the experiment!

Key Lesson: Building Causality in Experiment

Why are experiments considered the "gold standard"?

- **Random assignment** helps control for all confounding variables
  - Only difference (in expectation) across the different conditions is the treatment variable that you are manipulating
- This allows you to infer a **clean causal relationship** between X (treatment variable) and Y (dependent variable).
- In Marketing, we often conduct A/B testing for the data-driven decision making!

7. Project Management

Why Project Management Skills Matter

- What is managing project? Why important for NPD?
  - How different to other process management?
  - If project management fails?

- Managing Project Process
  - Defining and Organizing Projects
  - Constructing Project Networks
    - Defining the Work Breakdown Structure (WBS)
    - Diagramming the Network
    - Developing the schedule using critical paths

Project

- Unique, **one-time operations** designed to accomplish a specific set of objectives in a **limited time frame**
- Interrelated set of activities with a definite starting and ending point, which results in a unique outcome for a specific allocation of resources

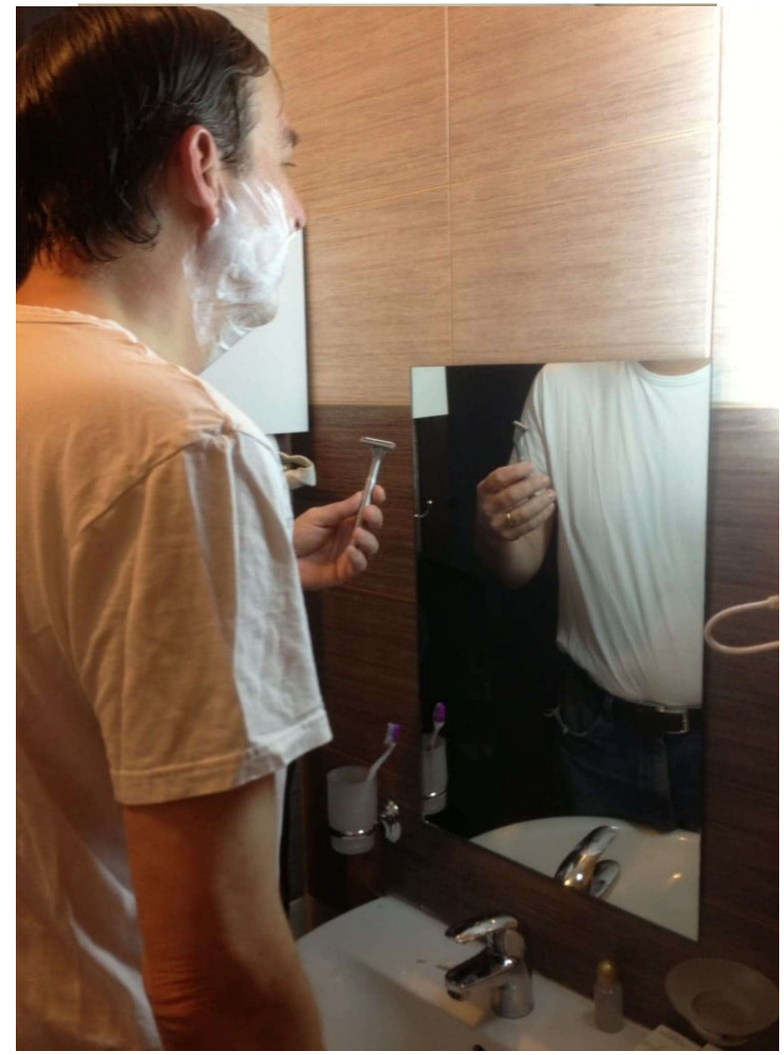
- Constructing a shopping complex
- Merging two companies
- Designing and running a political campaign
- Hosting the Olympic games
- Producing a motion picture of a music video
- **New product/service development**
- Designing advertising campaigns
- Reengineering a process
- Software development and many more…

Project Management Gone Wrong

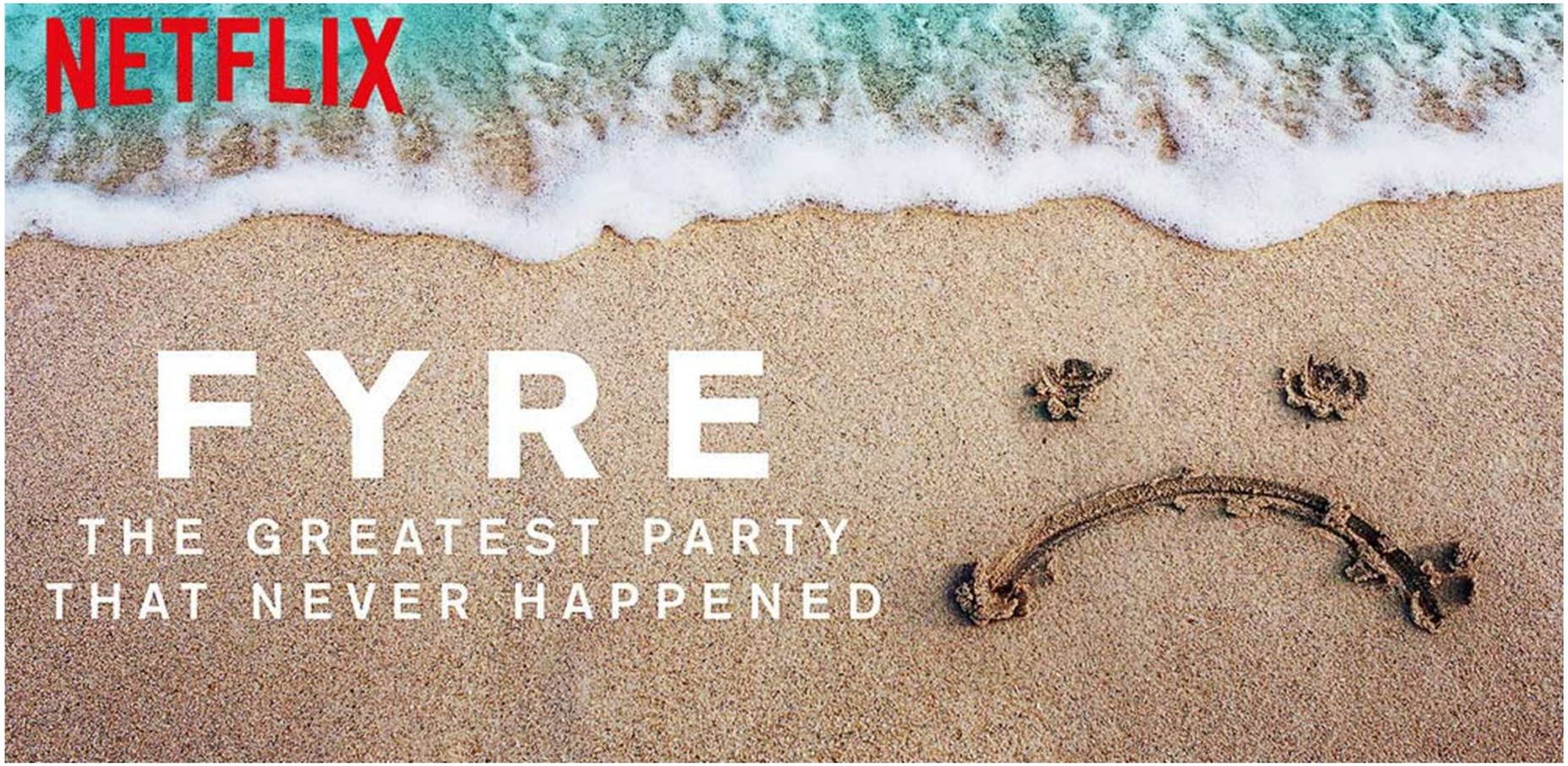
- Sochi 2014

Project Management Gone Wrong

- Sochi 2014



Project Management Gone Disastrous



Three Main Goals

1. Complete the project on time or earlier

2. Do not exceed the budget

3. Meet the specifications for customer satisfaction

Project Management

1. Defining and organizing projects
   - Defining the scope/goal of project
   - Selecting project manager
   - Recognizing organizational structure

Product manager should play the role of project manager!

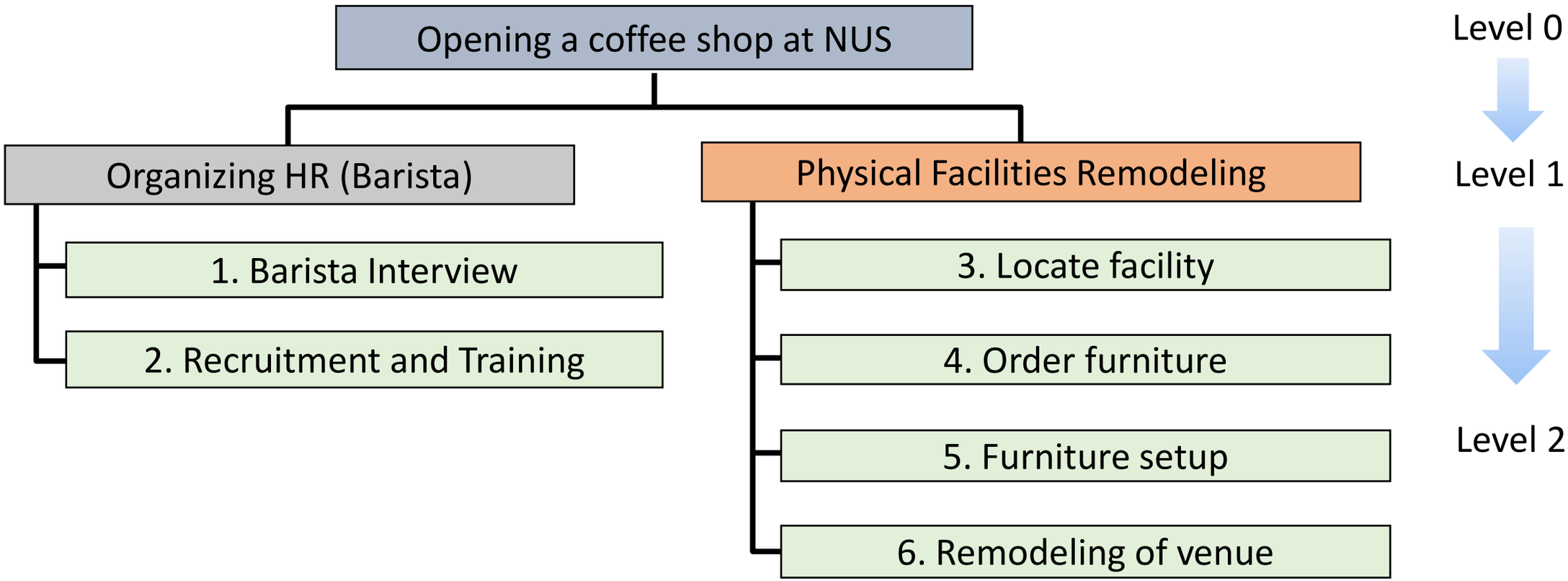
2. Constructing project networks
   - Recognize Work Breakdown Structure (WBS)
   - Diagramming the network
   - Developing the project schedule using critical paths
   - Analyzing cost-time trade-offs
   - Assessing and analyzing risks

Work Breakdown Structure (WBS)

- Work Breakdown Structure (WBS)
  - A statement of all work that must be completed
  - A hierarchical listing of what must be done during a project
  - Establishes a logical framework for identifying the required activities for the project

- Activity
  - The smallest unit of work effort consuming both time and resources that the project manager can schedule and control

WBS Example


Opening a coffee shop at NUS
Organizing HR (Barista)
Physical Facilities Remodeling
1. Barista Interview
2. Recruitment and Training
3. Locate facility
4. Order furniture
5. Furniture setup
6. Remodeling of venue
Level 0
Level 1
Level 2

Diagramming the Network

- Network diagram
  - A network planning method designed to depict the relationships between activities
  - consist of nodes (circles) and arcs (arrows)

- Precedence relationship
  - A relationship that determines a sequence for undertaking activities; it specifies that one activity cannot start until a preceding activity has been completed.

Diagramming the Network

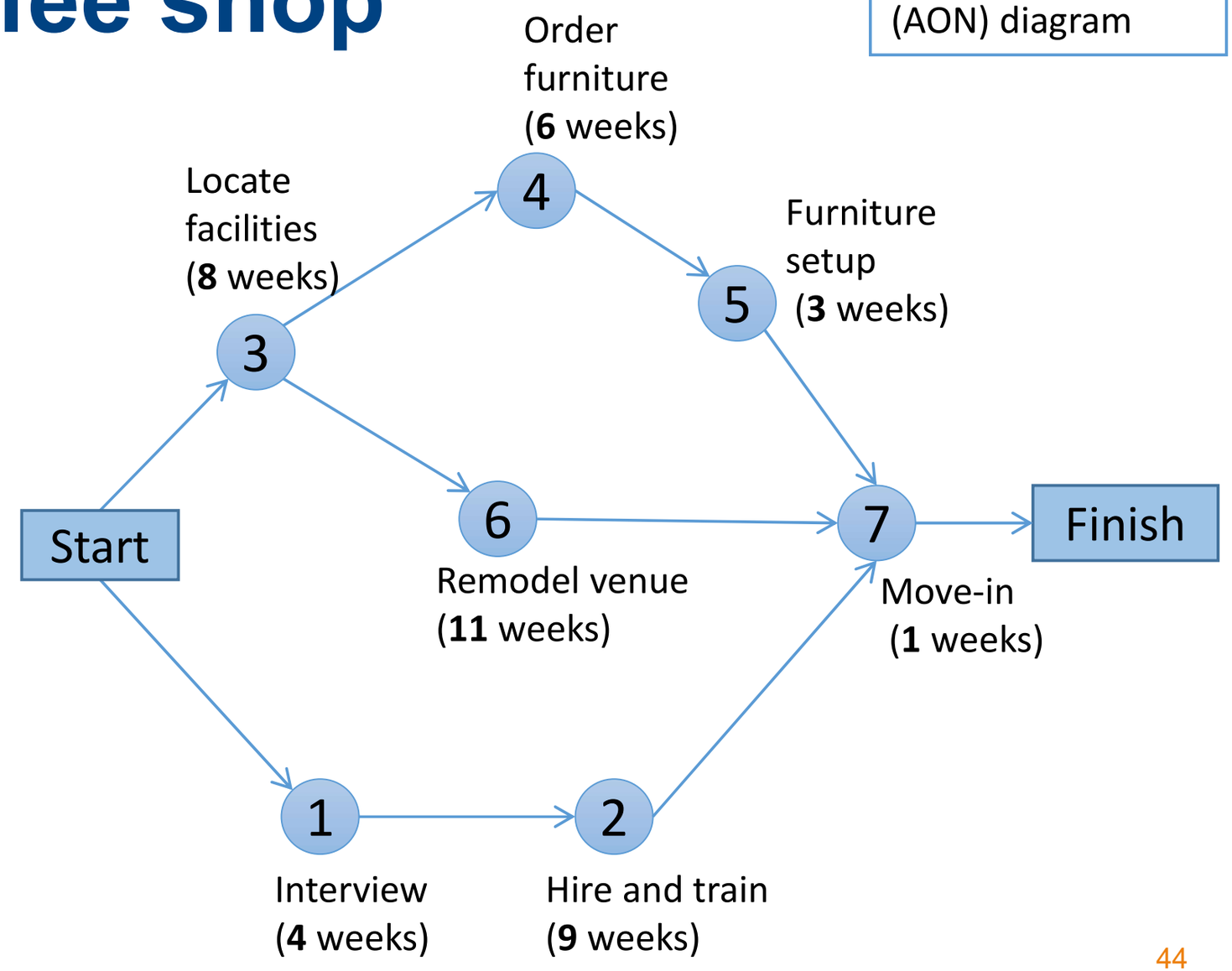
- **PERT** (Program Evaluation and Review Technique) and **CPM** (Critical Path Method) are widely used techniques for planning and coordinating large-scale projects. These are two different names for essentially the same technique.

Through PERT & CPM, managers can obtain:

- A graphical display of project activities.
- An estimate of how long the project will take.
- An indication of which activities are the most critical to timely project completion.
- An indication of how long any activity can be delayed without delaying the project.

Opening a coffee shop

Activities-On-Node (AON) diagram



Project Time Estimates:

a. Length of each path?

b. The critical path?

c. The expected length of the project?

d. The amount of slack time for each path?

Terms

- Activity: Project steps that consume resources and time. Activity is the smallest unit of work effort that the project manager can schedule/control.
- Path: A sequence of activities that leads from the starting node to the finishing node. (node – a connecting point)
- Critical Path: The path that takes longest completion time. Critical path determines expected project duration.
- Critical Activity: Activities on the critical path.
- Slack: Allowable slippage for a path, i.e., the difference between the length of a path and the length of the critical path. Path that are shorter than the critical path can experience some delays and still not affect the overall project completion time.

Practice Problem

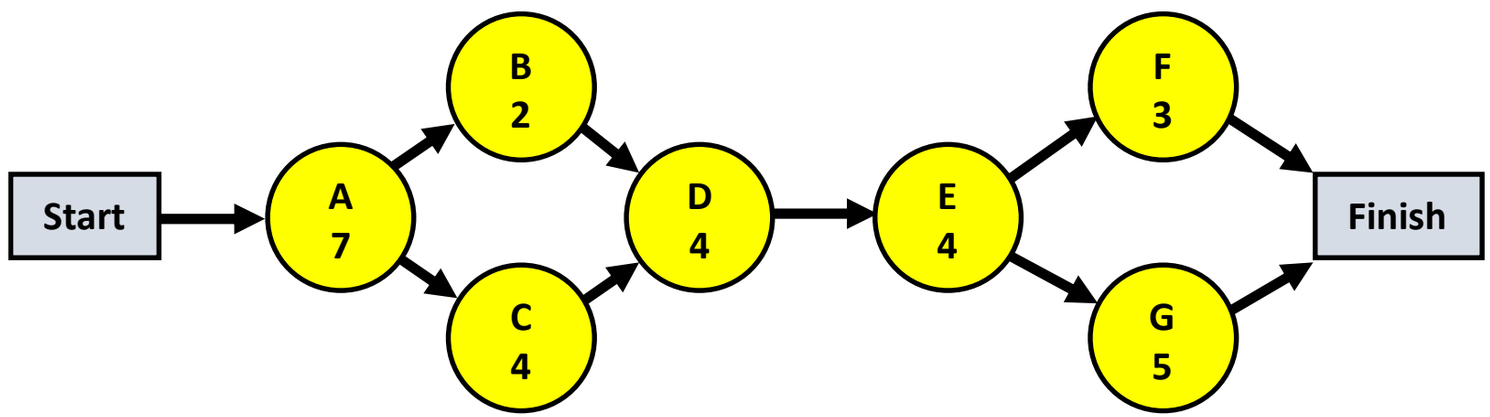
- Suppose the following information is known about a project

Activity	Activity Time (days)	Immediate Predecessor(s)
A	7	—
B	2	A
C	4	A
D	4	B, C
E	4	D
F	3	E
G	5	E

Draw the network diagram for this project and find the expected duration of the project!

Practice Problem

Activity	Activity Time (days)	Immediate Predecessor(s)
A	7	—
B	2	A
C	4	A
D	4	B, C
E	4	D
F	3	E
G	5	E

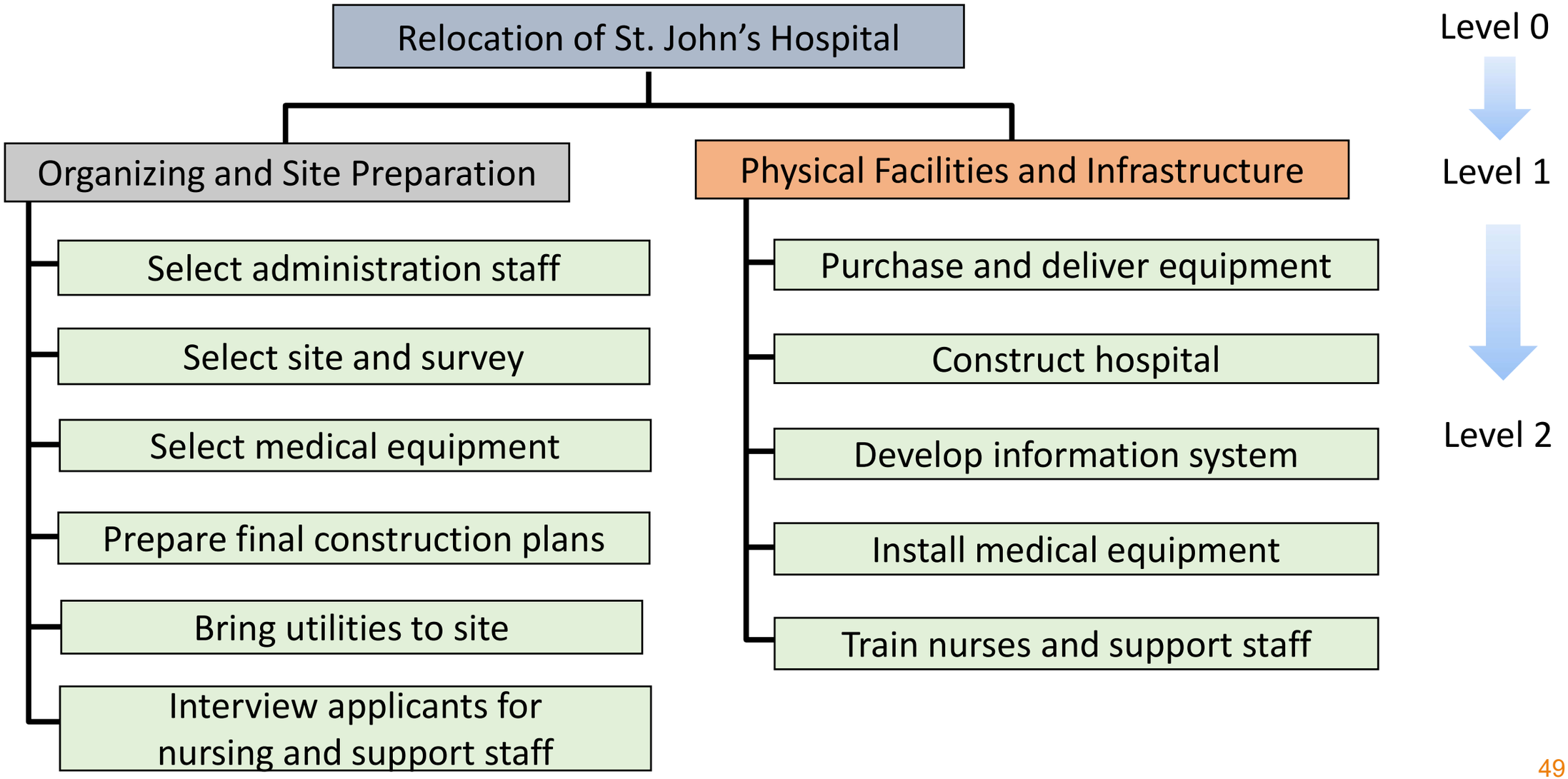


Remember from Today

- In quantitative market research:
  - ✓ Remember how statistical association can be different to causation
  - ✓ Remember how unincentivized survey methods are prone to biases
    - ✓ Experiment!
    - ✓ Incentivize subjects or consider alternatives!

- Project scheduling tool (PERT/CPM diagram)

WBS: Hospital Relocation Project



Example

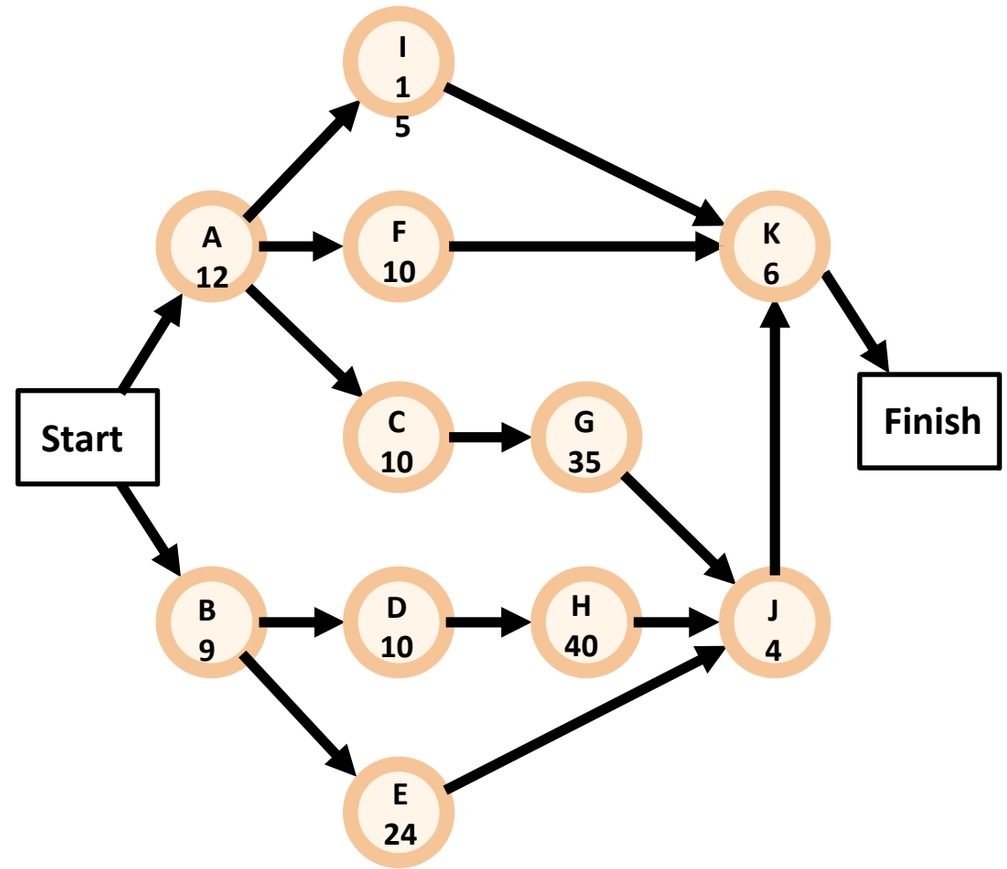
Activity	Immediate Predecessors	Activity Times (wks)	Responsibility
ST. JOHN'S HOSPITAL PROJECT			Kramer
START		0	
ORGANIZING and SITE PREPARATION			Stewart
A. Select administrative staff	Start	12	Johnson
B. Select site and survey	Start	9	Taylor
C. Select medical equipment	A	10	Adams
D. Prepare final construction plans	B	10	Taylor
E. Bring utilities to site	B	24	Burton
F. Interview applicants for nursing and support staff	A	10	Johnson
PHYSICAL FACILITIES and INFRASTRUCTURE			Walker
G. Purchase and deliver equipment	C	35	Sampson
H. Construct hospital	D	40	Casey
I. Develop information system	A	15	Murphy
J. Install medical equipment	E, G, H	4	Pike
K. Train nurses and support staff	F, I, J	6	Ashton
FINISH	K	0	

	Immediate Predecessor
A	—
B	—
C	A
D	B
E	B
F	A
G	C
H	D
I	A
J	E,G,H
K	F,I,J

Example

	Immediate Predecessor
A	—
B	—
C	A
D	B
E	B
F	A
G	C
H	D
I	A
J	E,G,H
K	F,I,J

- Network Diagram

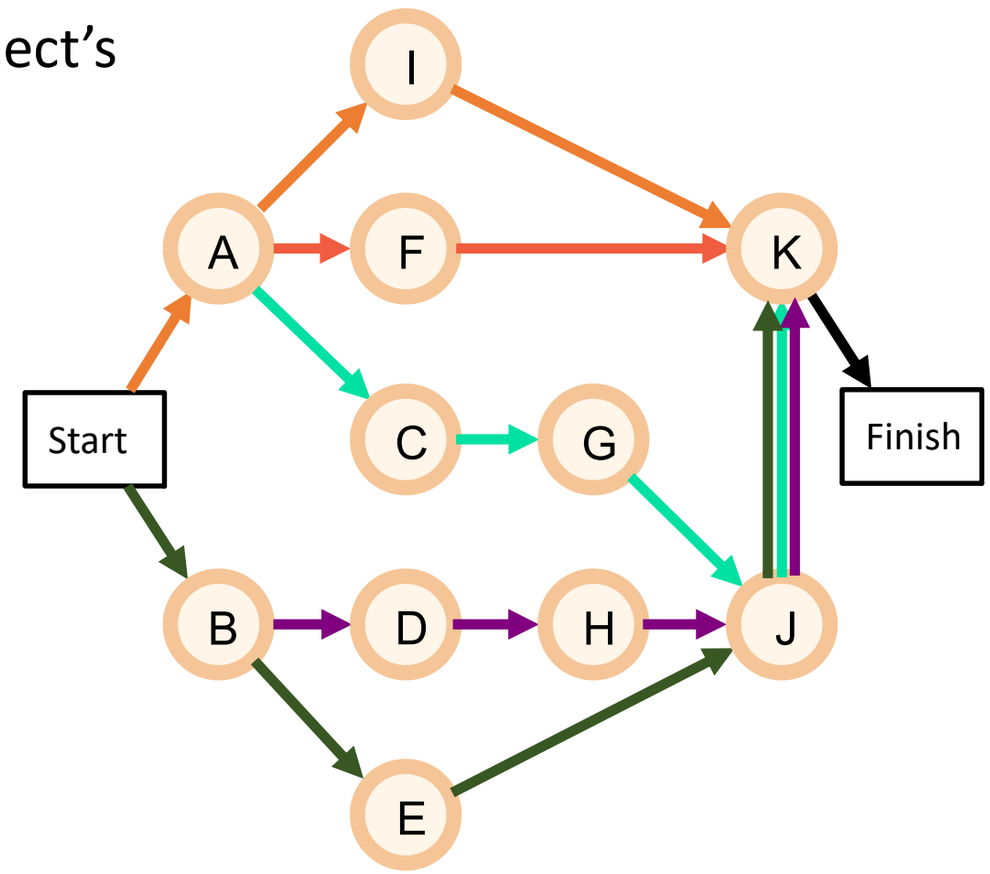


Example Problem

a. The critical path?
b. The expected length of the project?
c. Amount of slack time for each path?

Paths are the sequence of activities between a project's start and finish.

Path	Time (weeks)
A-I-K	33
A-F-K	28
A-C-G-J-K	67
B-D-H-J-K	69
B-E-J-K	43



Developing the Schedule

- Earliest start time (ES) - The earliest finish time of the immediately preceding activity.
- Earliest finish time (EF) - An activity's earliest start time plus its estimated duration (t).

  $EF = ES + t$

- Latest finish time (LF) - The latest start times of the activity that immediately follows.
- Latest start time (LS) - The latest finish time minus its estimated duration (t).

  $LS = LF – t$

- Activity Slack - The maximum length of time that an activity can be delayed without delaying the entire project

  Slack (S) = LS – ES (or, Slack = LF – EF)

Developing the Schedule

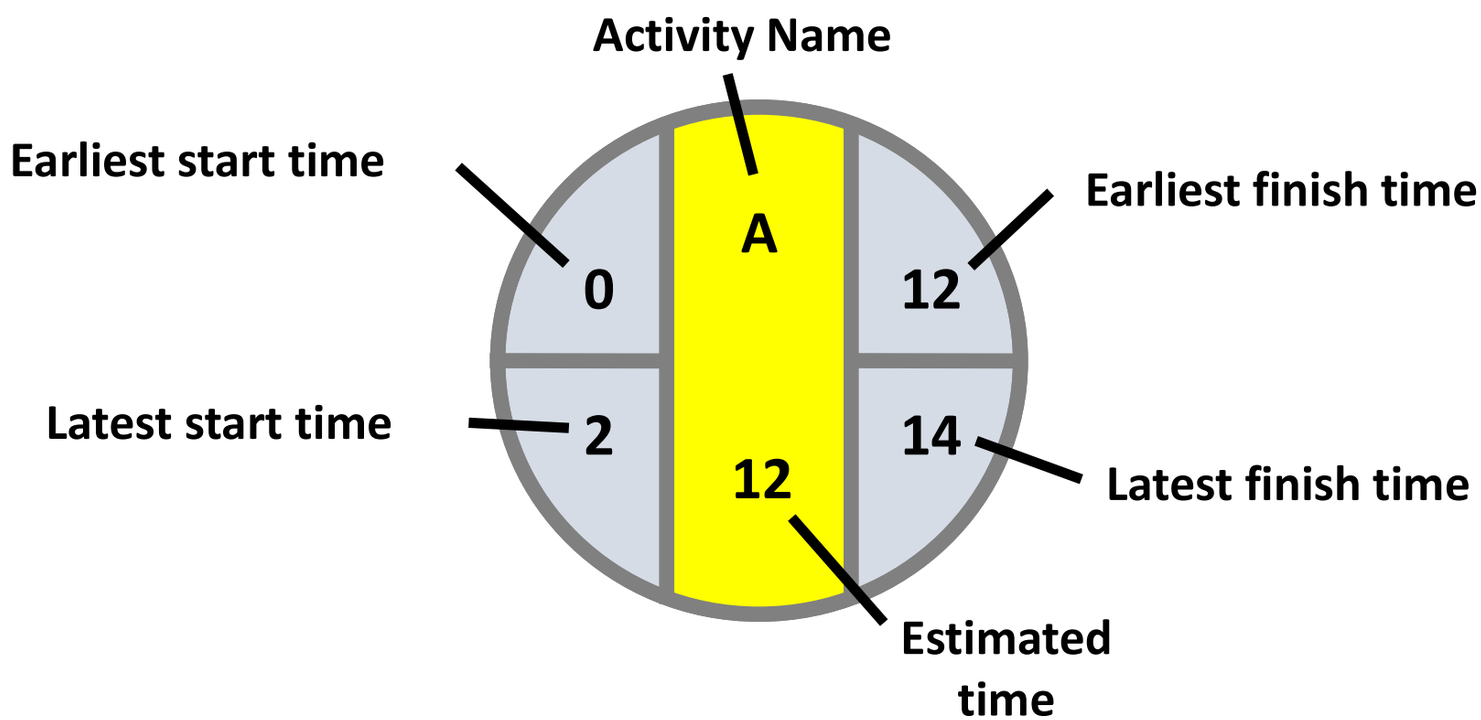


Figure 7.3

Example

Paths are the sequence of activities between a project's start and finish.

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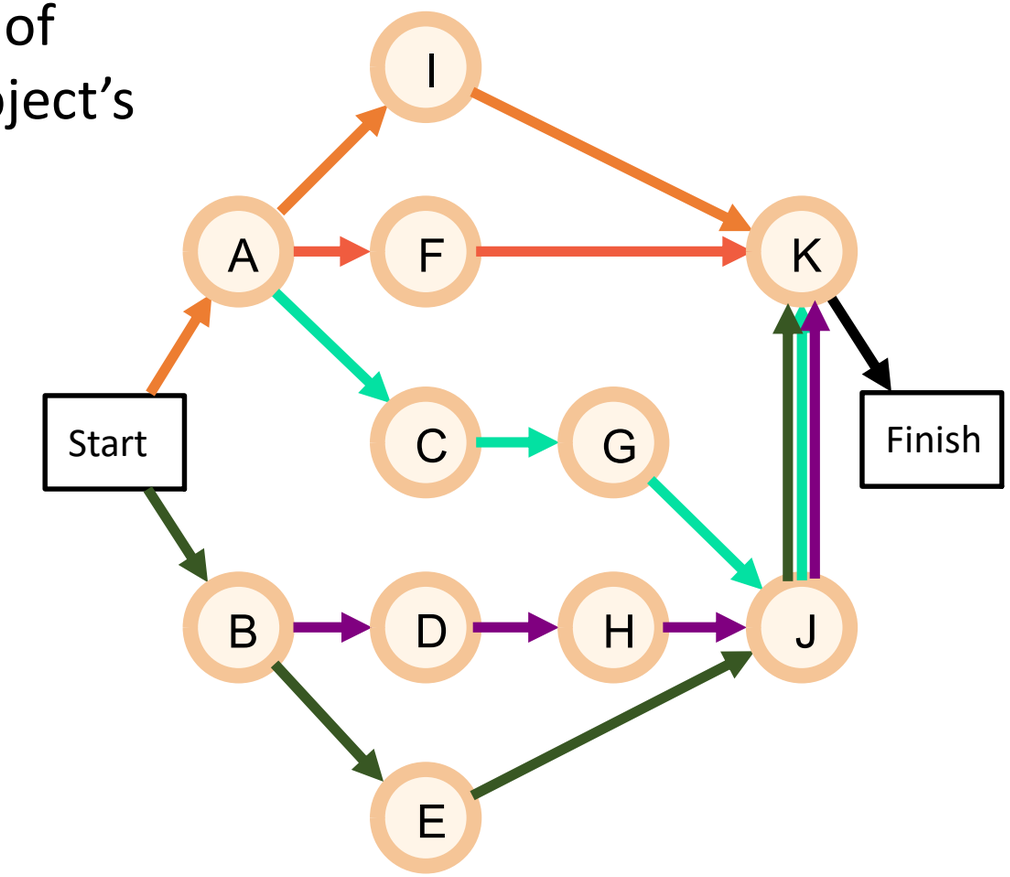
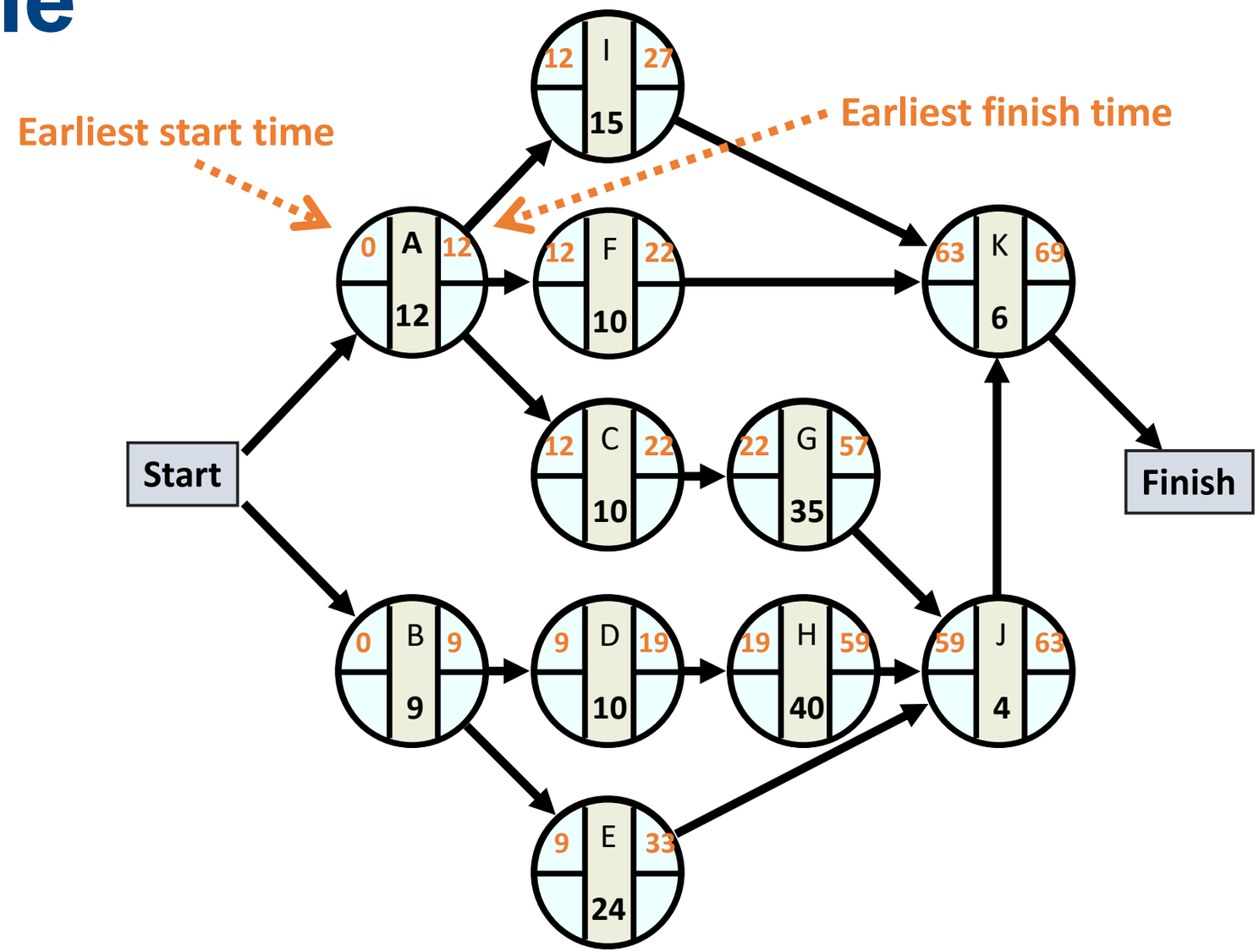
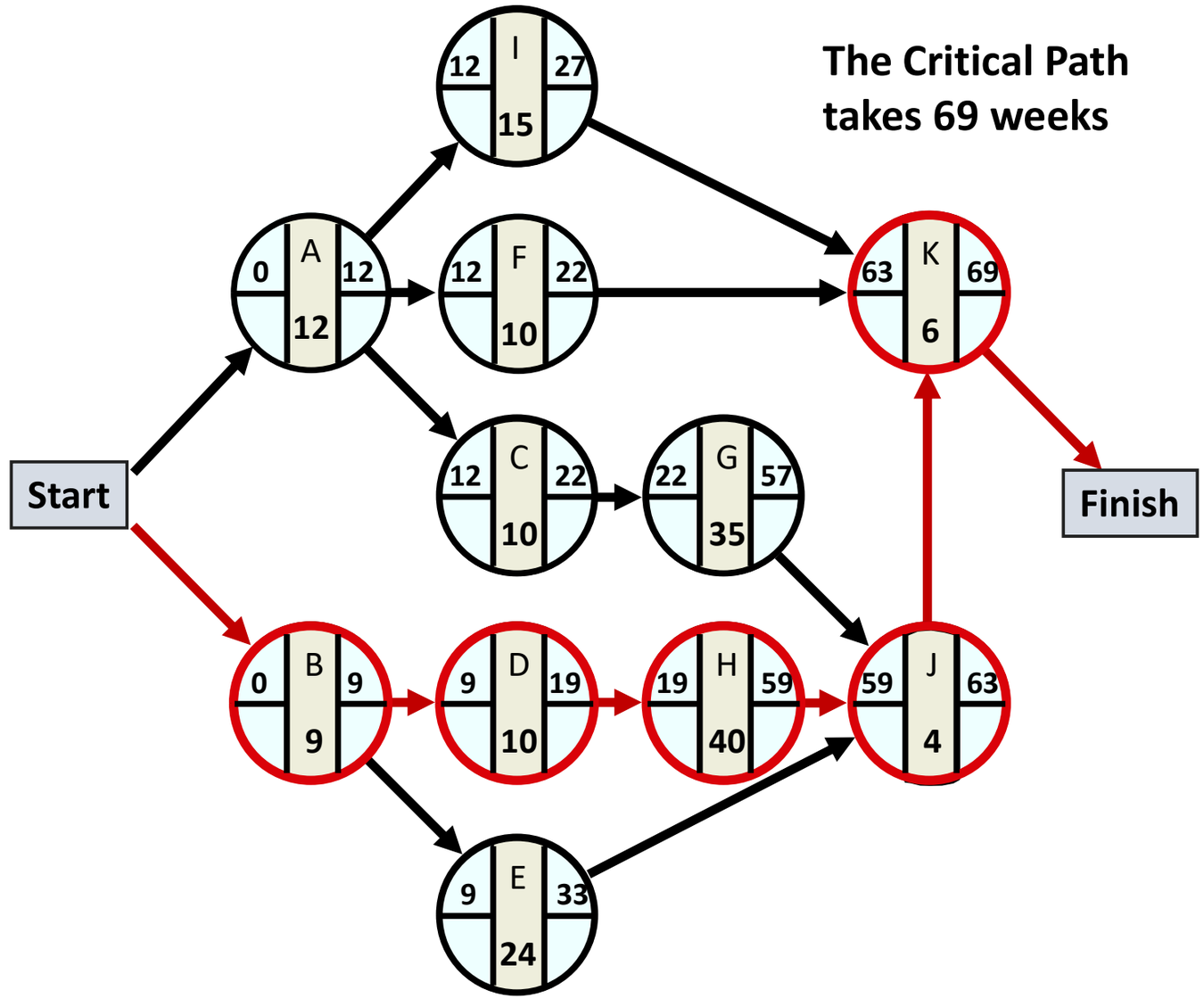
B-E-J-K	43



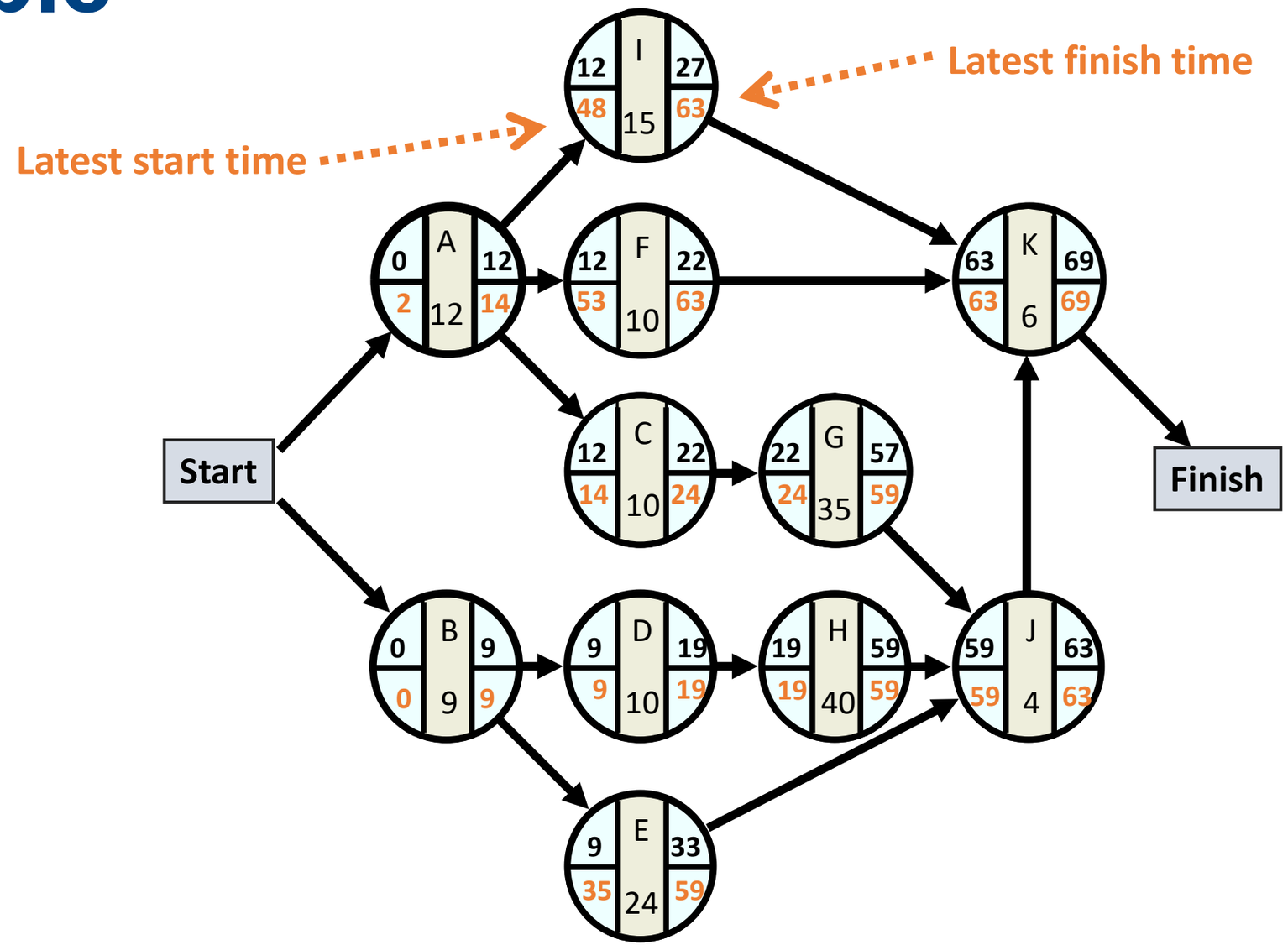
Example


Earliest start time
Earliest finish time
Start
Finish
0 A 12 12
12 I 27 15
12 F 22 10
12 C 22 10
22 G 57 35
0 B 9 9
9 D 19 10
19 H 59 40
9 E 33 24
59 J 63 4
63 K 69 6

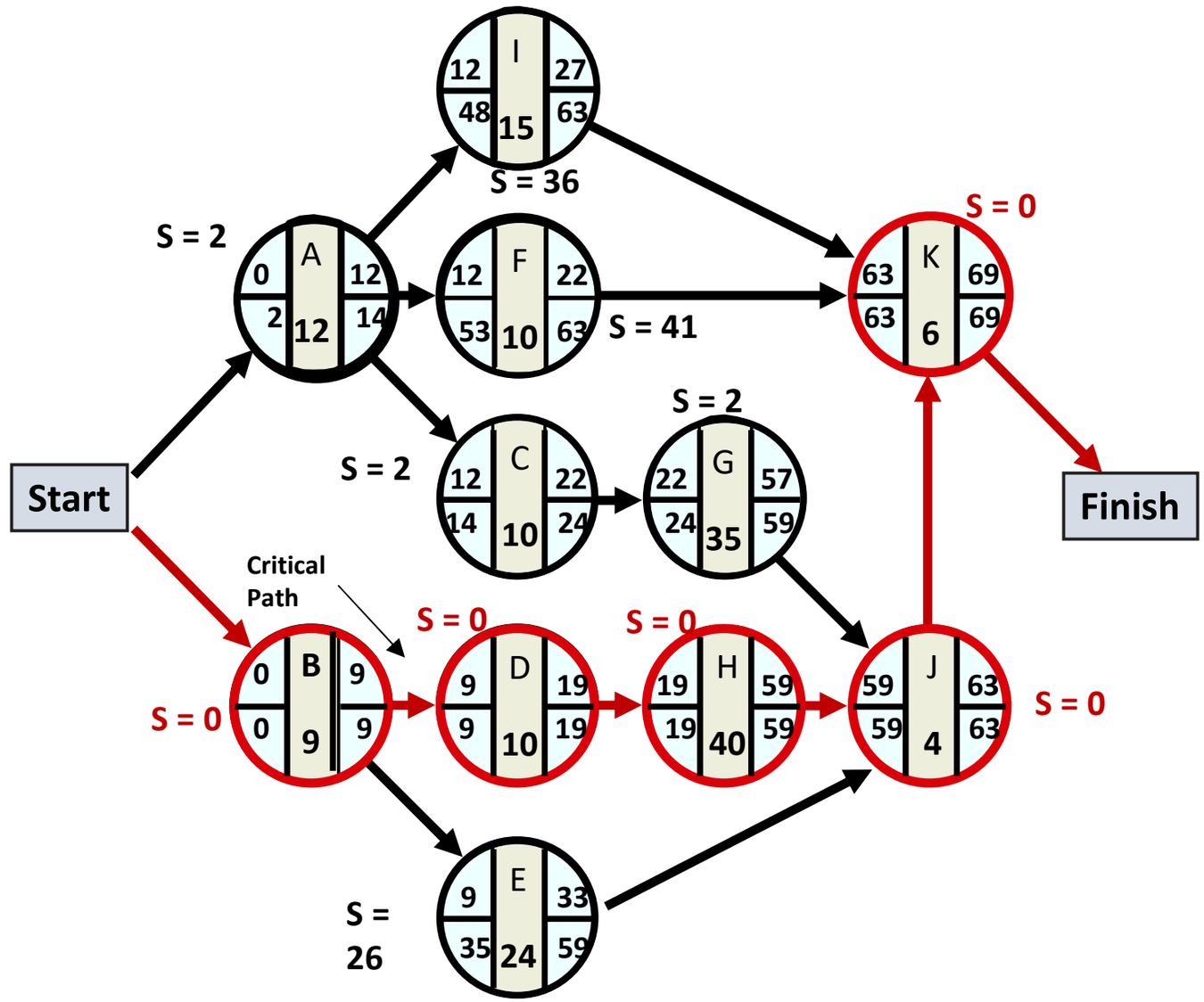
Example


The Critical Path takes 69 weeks
Start
Finish
A 0 12 12
B 0 9 9
C 12 22 10
D 9 19 10
E 9 33 24
F 12 22 10
G 22 57 35
H 19 59 40
I 12 27 15
J 59 63 4
K 63 69 6

Example


Latest finish time
Latest start time
Start
Finish
A 0 12 2 12 14
B 0 9 0 9 9
C 12 10 22 14 24
D 9 10 19 9 19
E 9 24 33 35 59
F 12 10 22 53 63
G 22 35 57 24 59
H 19 40 59 19 59
I 12 15 27 48 63
J 59 4 63 59 63
K 63 6 69 63 69

Example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2
12
12
14
S = 2
I
12
48
15
27
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F
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53
10
22
63
S = 41
C
12
14
10
22
24
S = 2
G
22
24
35
57
59
S = 2
K
63
63
6
69
69
S = 0
B
0
0
9
9
9
S = 0
Critical
Path
D
9
9
10
19
19
S = 0
H
19
19
40
59
59
S = 0
J
59
59
4
63
63
S = 0
E
9
35
24
33
59
S =
26

Analyzing Cost-Time Trade-Offs

- Project Crashing:
  Shortening (or expediting) some activities within a project to reduce overall project completion time and total project costs

- Determining the Minimum-Cost Schedule:
  1. Determine the project's critical path(s).
  2. Find the activity or activities on the critical path(s) with the lowest cost of crashing per week.
  3. Reduce the time for this activity until it cannot be further reduced or another path becomes critical
  4. Repeat this procedure until the increase in direct costs is larger than the savings generated by shortening the project.

Monitoring and Controlling Projects

- Projects have a *life cycle* that consists of four major phases

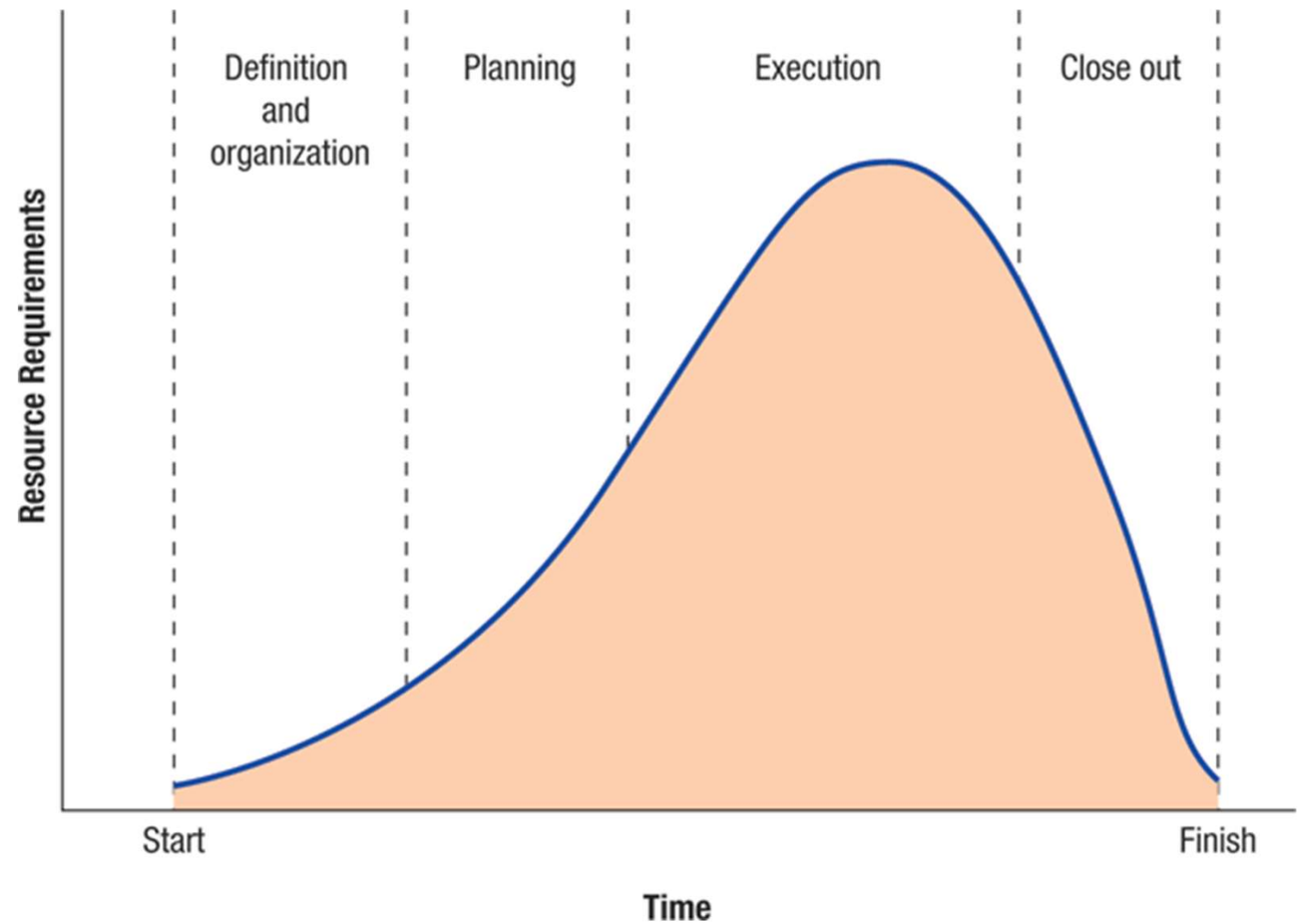

Figure 7.8

Monitoring and Controlling Projects

1. Definition and organization:
   a. Concept – at which point the organization recognizes the need for a project or responds to a request for a proposal.
   b. Feasibility Analysis – which examines the expected costs, benefits, and risks of undertaking the project.

2. Planning:
   - Spelling out the details of the work and providing estimates of the necessary human resources, time, and cost.

3. Execution:
   - During which the project itself is done. This phase often accounts for the majority of time and resources consumed by a project.

4. Close-out:
   - During which closure is achieved – writing final reports, completing remaining deliverables, reassigning personnel, dealing with leftover materials, equipment, etc.

THANK YOU